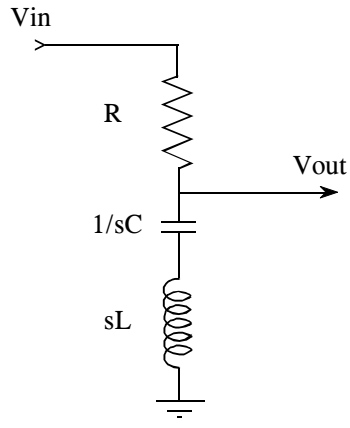


FIGURE 32-6
Notch filter analysis in the s-domain. The first step in this procedure is to replace the resistor, inductor & capacitor values with their s-domain equivalents.

So far, this is identical to the techniques of the last chapter, except for using *s* instead of $j\omega$. The difference between the two methods is what happens from this point on. This is as far as we can go with $j\omega$. We might graph the frequency response, or examining it in some other way; however, this is a mathematical dead end. In comparison, the interesting aspects of the Laplace transform have just begun. Finding $H(s)$ is the key to Laplace analysis; however, it must be expressed in a particular form to be useful. This requires the algebraic manipulation of the next two steps.

Step 3: Arrange $H(s)$ to be one polynomial over another. This makes the transfer function written as:

EQUATION 32-2
Transfer function in polynomial form.

$$H(s) = \frac{as^2 + bs + c}{\mathrm{a}s^2 + \mathrm{b}s + \mathrm{c}}$$

It is always possible to express the transfer function in this form *if* the system is controlled by differential equations. For example, the rectangular pulse shown in Fig. 32-3 is not the solution to a differential equation and its Laplace transform cannot be written in this way. In comparison, any electric circuit composed of resistors, capacitors, and inductors can be written in this form. For the *RLC* notch filter used in this example, the algebra shown in *step 2* has already placed the transfer function in the correct form, that is:

$$H(s) = \frac{as^2 + bs + c}{\mathrm{a}s^2 + \mathrm{b}s + \mathrm{c}} = \frac{Ls^2 + 1/C}{Ls^2 + Rs + 1/C}$$

where: $a = L$, $b = 0$, $c = 1/C$; and $\mathrm{a} = L$, $\mathrm{b} = R$, $\mathrm{c} = 1/C$

Step 4: Factor the numerator and denominator polynomials. That is, break the numerator and denominator polynomials into components that each contain